Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No: 9

Title: Exploratory data analysis on business datasets

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Exploratory data analysis on business datasets

Problem Statement :

You are a Data Analyst at a consulting firm. A client has provided a large business dataset and wants to understand key trends and patterns. Perform exploratory data analysis (EDA) to uncover meaningful business insights and identify opportunities for improvement.

.1. Introduction

In today's competitive business environment, organizations generate large amounts of data from sales transactions, customers, products, regions, and business operations. Analyzing this data helps organizations understand their current performance and make better business decisions.

Exploratory Data Analysis (EDA) is an important process used to examine datasets, identify trends and patterns, detect unusual observations, and discover relationships between different variables. Data visualization tools such as Microsoft Power BI make it easier to transform raw business data into meaningful and interactive dashboards.

This project performs exploratory data analysis on a **Superstore business dataset** using Microsoft Power BI. The analysis focuses on sales, profit, orders, quantity, product categories, regions, customer segments, and shipping methods. The objective is to identify important business trends and provide insights that can help management improve business performance.

2. Problem Statement

You are a Data Analyst at a consulting firm. A client has provided a large business dataset and wants to understand key trends and patterns. Perform exploratory data analysis (EDA) to uncover meaningful business insights and identify opportunities for improvement.

The client needs a dashboard that can transform the available business transaction data into useful information. The analysis should help identify high-performing and underperforming categories, regions, customer segments, and shipping methods while also examining overall sales and profitability.

3. Objective

The main objectives of this project are:

1. To perform exploratory data analysis on the Superstore business dataset.
2. To analyze overall sales and profit performance.
3. To identify sales trends across different years.
4. To compare sales and profitability across product categories and sub-categories.
5. To analyze business performance across different geographical regions.
6. To understand sales contribution from different customer segments.
7. To analyze sales distribution across different shipping modes.
8. To identify important business trends and patterns through interactive visualizations.
9. To identify areas where business performance can be improved.
10. To develop an interactive Power BI dashboard for business decision-making.

4. Dataset Description

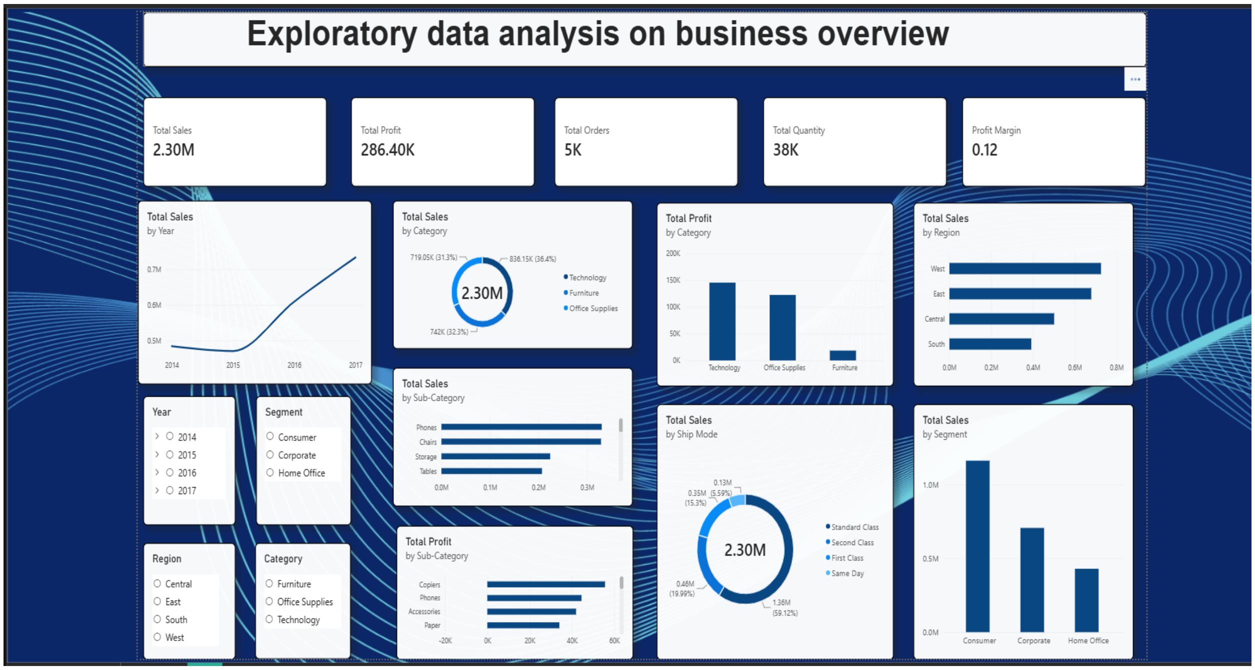
The project uses the **Sample Superstore dataset**, which contains transactional information about a retail business.

The dataset contains approximately **9,994 records and 21 attributes**.

Important attributes

Attribute	Description
Row ID	Unique identifier for each record
Order ID	Unique order identification number
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Ship Mode	Method used for shipping
Customer ID	Unique customer identifier
Customer Name	Name of the customer
Segment	Customer segment
Country	Country of the customer
City	Customer city
State	Customer state
Region	Business region
Product ID	Unique product identifier
Category	Main product category
Sub-Category	Product sub-category
Product Name	Name of the product
Sales	Revenue generated from the sale
Quantity	Number of products sold
Discount	Discount offered
Profit	Profit generated from the transaction

5. Dashboard Design



6. Tools and Technologies Used

Microsoft Power BI

Microsoft Power BI was used to:

- Import the dataset
- Clean and prepare the data
- Create calculated measures using DAX
- Develop interactive visualizations
- Analyze business trends
- Create an interactive dashboard

Data Analysis Expressions (DAX) was used to calculate important business metrics such as total sales, total profit, total orders, total quantity, profit margin, and average order value.

Total Sales = SUM('Sample - Superstore'[Sales])

Total Profit = SUM('Sample - Superstore'[Profit])

Total Orders = DISTINCTCOUNT('Sample - Superstore'[Order ID])

Total Quantity = SUM('Sample - Superstore'[Quantity])

Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)

Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)

7. Data Preprocessing

Before creating the dashboard, the dataset was checked and prepared for analysis.

The following preprocessing activities were performed:

7.1 Data Type Verification

The data types of the columns were checked to ensure that:

- Order Date was treated as a date.
- Ship Date was treated as a date.
- Sales was treated as a numerical value.
- Profit was treated as a numerical value.
- Quantity was treated as a whole number.
- Discount was treated as a decimal value.

7.2 Data Quality Check

The dataset was examined for:

- Missing values
- Incorrect data types
- Duplicate records
- Invalid numerical values
- Inconsistent categorical values

The prepared dataset was then used to create the Power BI dashboard.

8. Dashboard Design

The Power BI dashboard was designed to provide a comprehensive overview of business performance.

The dashboard contains KPI cards, charts, slicers, and interactive visualizations.

Main KPIs

The dashboard displays:

- **Total Sales:** approximately **2.30M**
- **Total Profit:** approximately **286.40K**
- **Total Orders:** approximately **5K**
- **Total Quantity:** approximately **38K**
- **Profit Margin:** approximately **12%**

These KPIs provide a quick overview of the organization's overall performance.

9. Dashboard Analysis and Interpretation

9.1 Sales Trend by Year

The line chart shows the change in total sales across the years.

From the dashboard, sales remain relatively stable during **2014 and 2015**, followed by a significant increase in **2016 and 2017**.

The highest sales level is observed toward the end of **2017**.

Insight

The increasing sales trend in the later years indicates positive business growth. The organization could investigate the factors contributing to the strong growth during 2016–2017 and use those strategies to maintain future growth.

10. Sales by Category

The dashboard divides sales into three major categories:

- Technology
- Furniture
- Office Supplies

According to the dashboard:

Technology contributes the highest sales among the three categories.

Insight

Technology is the strongest sales-generating category, contributing approximately 36% of total sales. Furniture and Office Supplies also contribute significant portions of revenue, indicating that the business has a relatively diversified product portfolio.

11. Profit by Category

The profit chart shows the profitability of each major product category.

From the dashboard:

- Technology generates the highest profit.

- Office Supplies generates the second-highest profit.
- Furniture generates the lowest profit among the three categories.

Insight

Technology is the strongest category in terms of both sales and profit. Furniture, despite contributing a substantial amount of sales, generates considerably less profit, indicating that its pricing, discount levels, or product costs may require further investigation.

12. Sales by Region

The dashboard compares sales across four regions:

- West
- East
- Central
- South

The **West region** has the highest sales, followed by the **East**, **Central**, and **South** regions.

Insight

The West region is the strongest market in terms of sales, while the South region has comparatively lower sales. Management can study the successful strategies used in the West and consider applying suitable strategies to improve performance in weaker regions.

13. Sales by Sub-Category

The sub-category analysis identifies the products contributing most to overall sales.

The dashboard shows strong sales from sub-categories such as:

- Phones
- Chairs
- Storage
- Tables

Insight

Phones and Chairs are among the leading sub-categories in terms of sales. These products can be prioritized for inventory planning and marketing campaigns because they represent important sources of revenue.

14. Profit by Sub-Category

The profit analysis provides a different perspective from sales analysis.

The dashboard highlights sub-categories such as:

- Copiers
- Phones
- Accessories
- Paper

Insight

The most profitable sub-categories are not necessarily identical to the highest-sales sub-categories. This demonstrates why businesses should evaluate both revenue and profitability before making product decisions.

15. Sales by Customer Segment

The dashboard compares three customer segments:

- Consumer
- Corporate
- Home Office

The **Consumer segment** generates the highest sales, followed by Corporate and Home Office.

Insight

Consumers are the largest contributors to sales. This indicates that the business should maintain strong consumer-focused marketing strategies while also exploring opportunities to increase sales from Corporate and Home Office customers.

16. Sales by Ship Mode

The dashboard analyzes sales across different shipping methods:

- Standard Class
- Second Class
- First Class
- Same Day

Standard Class accounts for the largest share of sales.

Insight

> Standard Class is the most frequently associated shipping method with sales. This may indicate that customers generally prefer economical shipping options. However, premium shipping options can still be evaluated to determine whether they provide additional revenue opportunities.

17. Key Business Insights

Based on the exploratory analysis, the following major insights were identified:

1. Strong overall business performance

The business generates approximately **$2.30 million in sales** and **$286.40K in profit**, indicating positive overall profitability.

2. Technology is the leading category

Technology has the highest sales contribution and also generates the highest profit among the major categories.

3. Sales are growing over time

The sales trend shows significant growth during the later years, particularly from 2016 to 2017.

4. West is the strongest region

The West region records the highest sales, while the South region has comparatively lower sales.

5. Consumer customers dominate

The Consumer segment contributes the largest share of sales.

6. Furniture requires further investigation

Furniture generates substantial sales but comparatively lower profit, suggesting that its costs, pricing, or discount strategy should be examined.

7. Product-level analysis is important

High sales do not necessarily mean high profitability. Therefore, management should evaluate both sales and profit when making product decisions.

18. Opportunities for Improvement

Based on the analysis, several opportunities can be considered.

18.1 Improve Furniture Profitability

Management should investigate:

- Product costs
- Discounts
- Pricing
- Shipping costs

for Furniture products.

18.2 Focus on High-Performing Products

Products and sub-categories generating strong sales and profit can receive additional marketing and inventory support.

18.3 Improve Regional Performance

The company can analyze the strategies used in high-performing regions such as the West and identify ways to improve weaker regions.

18.4 Strengthen Consumer Marketing

Since the Consumer segment generates the highest sales, targeted promotions and loyalty programs can be used to retain and increase customer value.

18.5 Optimize Shipping

Since Standard Class accounts for the largest share of sales, the company can optimize inventory and logistics around this shipping method while evaluating the profitability of faster shipping options.

19. Conclusion

The exploratory data analysis performed using Microsoft Power BI successfully transformed the Superstore transactional dataset into meaningful business insights.

The dashboard provides a comprehensive view of sales, profit, orders, quantity, categories, sub-categories, regions, customer segments, and shipping methods. The analysis shows that

Technology is the strongest product category, the West is the leading region, and Consumers are the largest customer segment. The sales trend also indicates strong business growth during the later years.

The analysis further demonstrates that sales alone cannot be used to evaluate business performance. Profitability must also be considered, particularly when comparing product categories and sub-categories. Areas such as Furniture and lower-performing regions can be investigated further to identify opportunities for improvement.

Overall, the Power BI dashboard provides management with an interactive tool for understanding business performance, identifying trends, and supporting data-driven decision-making.

GITHUB LINK:

https://github.com/prajwaldotcom/PowerBi-DashBoards/tree/main/week%209-expwriment